

Volatility forecasting with GARCH model

Joshua Dinn

May 2025

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1 Introduction

Volatility modeling is a central theme in financial econometrics, with extensive implications for risk management, asset pricing, and strategic investment decisions. The S&P 500 index, representing a broad spectrum of the U.S. equities market, serves as an ideal candidate for investigating market volatility due to its comprehensive coverage and frequent analysis. In this report, we apply the Generalized Autoregressive Conditional Heteroskedasticity (GARCH) model to examine the time-varying volatility inherent in the S&P 500 index, aiming to provide insights into its dynamic risk profiles.

Traditional volatility measures assume constancy over time, a presumption often challenged by empirical evidence of volatility clustering—wherein periods of heightened or diminished volatility recur continuously in financial markets. The GARCH modeling framework addresses these complexities by allowing past variances and squared residuals to forecast future volatility, offering a nuanced understanding of market dynamics.

Through empirical analysis on historical data from the S&P 500, this study explores various GARCH configurations, identifying the GARCH(1,1) model as an optimal choice based on the Akaike Information Criterion (AIC) and Bayesian Information Criterion (BIC). Diagnostic tests including the Ljung-Box and ARCH LM tests further evaluate the model's residuals, revealing both its strengths and areas for refinement.

The findings suggest that while the GARCH(1,1) model adequately captures broad volatility patterns in the S&P 500 during the study period, significant autocorrelation and remaining ARCH effects in the residuals highlight the potential for model enhancement. These insights contribute to existing literature by demonstrating the practicality of GARCH models in volatility forecasting and emphasize the necessity for continuous refinement in model specifications to account for evolving market conditions.

2 Theoretical Framework

Volatility modeling plays a crucial role in deciphering the complexities of financial markets. The Generalized Autoregressive Conditional Heteroskedasticity (GARCH) model, introduced by Bollerslev1986, builds upon the foundations of the Autoregressive Conditional Heteroskedasticity (ARCH) model formulated by Engle1982. These models are designed to capture the dynamic, time-varying nature of market volatility and offer enhanced forecasting capabilities. One notable phenomenon these models address is volatility clustering, where periods of high volatility are often succeeded by similar high-volatility phases.

2.1 GARCH Model

The GARCH model further refines the ARCH framework by integrating lagged conditional variances into its formulation. This approach allows the model to account for past volatility patterns, improving its predictive accuracy. Specifically, a GARCH(p,q) model is characterized by:

$$\epsilon_t = \sigma_t z_t, \quad \sigma_t^2 = \alpha_0 + \sum_{i=1}^q \alpha_i \epsilon_{t-i}^2 + \sum_{j=1}^p \beta_j \sigma_{t-j}^2, \quad (1)$$

where σ_t^2 is the conditional variance, and α_i and β_j are non-negative parameters that ensure the positivity of the conditional variance. Typically, the GARCH(1,1) configuration is used due to its parsimony and effectiveness in capturing volatility for many financial time series. The GARCH(1,1) model specifically simplifies to:

$$\sigma_t^2 = \alpha_0 + \alpha_1 \epsilon_{t-1}^2 + \beta_1 \sigma_{t-1}^2. \quad (2)$$

2.2 Model Estimation

To implement a GARCH model, parameter estimation often relies on the Maximum Likelihood Estimation (MLE) method. The log-likelihood function for a sample of T observations

is given by:

$$\mathcal{L}(\Theta) = -\frac{1}{2} \sum_{t=1}^T \left(\log(2\pi) + \log(\sigma_t^2) + \frac{\epsilon_t^2}{\sigma_t^2} \right), \quad (3)$$

where Θ represents the vector of parameters α_0 , α_i , and β_j . Model selection criteria such as the Akaike Information Criterion (AIC) and Bayesian Information Criterion (BIC) can guide choices between different GARCH configurations.

2.3 Diagnostic Tests

To ensure the adequacy and robustness of volatility models, diagnostic tests are employed to analyze residuals for potential autocorrelation or lingering effects of conditional heteroskedasticity. Two commonly used tests in this context are the Ljung-Box test and the ARCH LM test.

2.3.1 Ljung-Box Test

The Ljung-Box test is used to detect autocorrelation in time series data, specifically in the residuals of fitted models. It assesses whether any group of autocorrelations up to a certain lag is significantly different from zero. The test statistic is given by:

$$Q = n(n+2) \sum_{k=1}^m \frac{\hat{\rho}_k^2}{n-k}, \quad (4)$$

where n is the number of observations, m is the number of lags being tested, and $\hat{\rho}_k$ represents the autocorrelation at lag k . The Q statistic follows a chi-square distribution with m degrees of freedom, and the null hypothesis states that there is no autocorrelation. A significant result indicates the presence of autocorrelation, suggesting the need for model refinement.

2.3.2 ARCH LM Test

The ARCH LM (Lagrange Multiplier) test is specifically designed to detect remaining ARCH effects in a time series, indicating conditional heteroskedasticity. The test involves regressing the squared residuals on a constant and their lagged values up to a pre-specified number of lags K . The test statistic is:

$$LM = nR^2, \quad (5)$$

where n is the number of observations, and R^2 is the coefficient of determination from the regression. Similar to the Ljung-Box test, the ARCH LM statistic follows a chi-square distribution with K degrees of freedom. A significant result suggests the presence of ARCH effects, indicating that the volatility model may require additional terms or a higher order specification.

Through these diagnostic tests, researchers can assess the sufficiency of the volatility model and make adjustments as necessary to improve its forecasting performance and structural integrity.

2.4 GARCH Model Forecasting

GARCH models provide dynamic forecasts of conditional variance by utilizing historical information embedded in lagged terms. For a GARCH(1,1) model, volatility forecasting is executed as follows:

$$\sigma_{t+h}^2 = \alpha_0 + \alpha_1 \epsilon_t^2 + \beta_1 \sigma_t^2, \quad (6)$$

where h denotes the forecast horizon. This formulation recursively extends the conditional variance, allowing for volatility predictions across multiple periods.

3 Data Description

This study analyzes the daily log returns of the S&P 500 index from 2015 to 2025. Log returns provide a continuous measure of asset price changes, ideal for volatility analysis.

Figure 1 shows the log returns, highlighting significant volatility, especially around 2020 due to market disruptions.

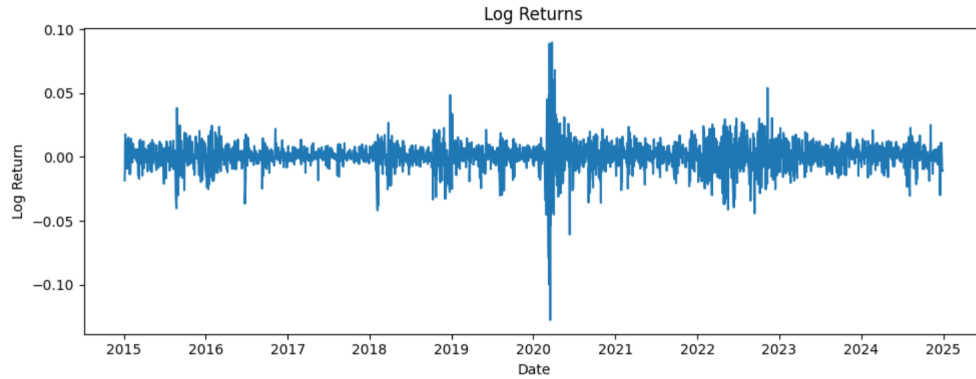


Figure 1: Log Returns of the S&P 500 Index (2015-2025)

Figure 2 presents squared log returns, emphasizing volatility spikes crucial for GARCH model forecasting.

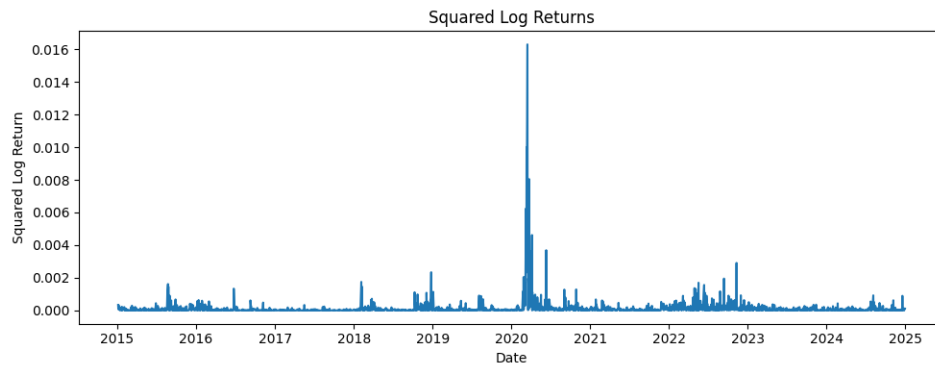


Figure 2: Squared Log Returns of the S&P 500 Index (2015-2025)

These figures effectively set the stage for applying GARCH models to forecast volatility, revealing the S&P 500's dynamic risk patterns.

4 Empirical analysis

Model estimation was carried out using Maximum Likelihood Estimation (MLE). The GARCH model was applied with a constant mean assumption. The estimated coefficients for the mean model are provided in Table 1. The mean coefficient μ is statistically significant, confirming that the chosen model effectively captures the average dynamics within the dataset.

Table 1: Mean Model Coefficients

	Coef	Std Err	t	P> t
μ	8.3305e-04	9.353e-07	890.649	0.000

The volatility model coefficients, as shown in Table 2, reveal that the constant term ω and parameters $\alpha[1]$ and $\beta[1]$ are all statistically significant. These parameters indicate that both past squared residuals and past variances contribute considerably to predicting current volatility.

Table 2: Volatility Model Coefficients

	Coef	Std Err	t	P> t
ω	2.5400e-06	1.273e-11	199500	0.000
$\alpha[1]$	0.2000	3.174e-02	6.302	2.941e-10
$\beta[1]$	0.7800	2.660e-02	29.319	5.869e-189

Table 3 summarizes the Ljung-Box test results, evaluating autocorrelation in model residuals across multiple lags. P-values indicate statistically significant autocorrelation, suggesting that additional model refinement could enhance the fit.

Table 3: Diagnostic Tests: Ljung-Box Test Results

Lag	LB Statistic	P-value
1	45.739720	1.350562e-11
2	61.293420	4.901205e-14
3	62.113169	2.078065e-13

Additionally, the ARCH LM Test results portrayed in Table 4 further assess the presence of heteroskedastic effects in the residuals. The highly significant p-value demonstrates the presence of remaining ARCH effects, warranting consideration of model adjustments or extensions.

Table 4: ARCH LM Test Results

Statistic	Value
LM Stat	971.86
p-value	2.15e-202

The forecasted volatility is expressed through predicted variances, extending the conditional variance estimates 100 days into the future. Figure 3 illustrates this forecast, showcasing the trend and expected changes over the specified horizon.

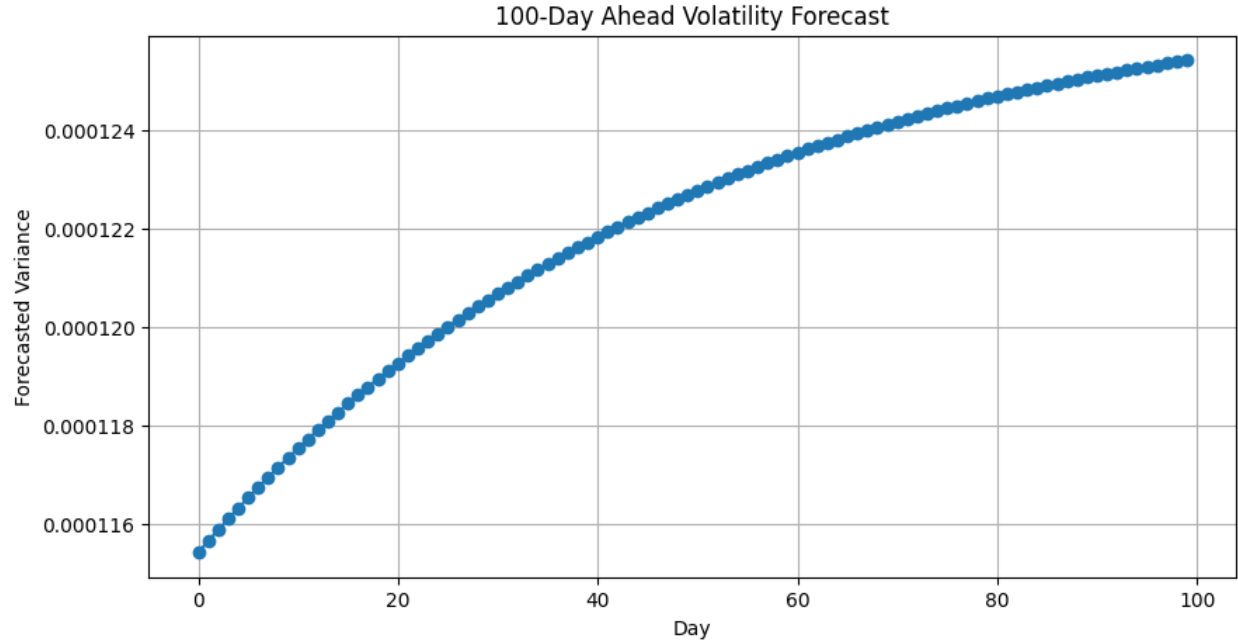


Figure 3: 100-Day Ahead Volatility Forecast

The trajectory of the forecasted variance shows a gradual increase in anticipated volatility. This upward trend suggests that the model expects volatility to rise incrementally over the coming 100 days, possibly due to the persistence captured by the GARCH parameters.

5 Discussion and Conclusion

The analysis of the S&P 500 index using the GARCH model provides valuable insights into the behavior of market volatility over the decade spanning 2015 to 2025. The study successfully illustrates the model's ability to capture significant volatility patterns characterized by volatility clustering.

The GARCH(1,1) model, identified as optimal through criteria such as AIC and BIC, proved effective in estimating both the mean and volatility dynamics of the dataset. This model's parameters revealed a considerable influence of past variances and residuals on current volatility levels, reinforcing the importance of historical data in financial forecasting.

Diagnostic tests, including the Ljung-Box and ARCH LM tests, highlighted areas for

potential refinement. The presence of significant autocorrelation and remaining ARCH effects suggests that extending model specifications could enhance prediction accuracy. Such extensions might include higher-order GARCH models or the incorporation of exogenous variables to account for external market influences.

The 100-day ahead volatility forecast predicts an incremental rise in market volatility. This foresight allows market participants to adjust their strategies, optimizing asset allocation and hedging against potential risk increases. Forecasts serve as vital tools for risk management, enabling proactive responses to evolving market conditions.

In conclusion, while the GARCH model offers a robust framework for volatility modeling, ongoing refinement and adaptation to market changes remain imperative. Future research may build upon these findings by exploring alternative volatility models or incorporating additional market variables to enhance forecast precision and reliability.